

Learning Objectives

- After completing this chapter, students will be able to;
 - Understand the meaning of Research
 - Understand the motives of doing research
 - Identify the types of research
 - State the objective of research
 - Differentiate b/n Research methods and Methodology

An Introduction

- Research is most frequently used terminology in different academic and business institutions.
- As per dictionary definition , the word research is composed of **two syllables**, **re** and **search**.
- **Re-means 'Again', 'a new or over again'**
- **Search-** means 'to examine closely and carefully', 'to test and try', or 'to probe'.
- Therefore, **research means 'to observe the phenomena again and again from different dimensions'**.
- It is a careful, systematic, patient study and investigation in some field of knowledge, undertaken to establish facts or principles.

Scientific Meaning of Research

- Research can mean different things to different people.
- It is an **essential tool** for understanding the events and structures of the social world.
- Where there is **a problem** there is always a **research**.
- Research commonly refers to a **search for knowledge**.
- Some people consider research as a movement from **known** to **unknown**.
- According to Herz, Research is the **application of human intelligence** in a **systematic manner** to a problem whose solution is **not immediately available**.

- **Woody** on the other hand defined research as an activity comprises of **defining** and **redefining** problems, **formulating hypothesis** or **suggested solutions**; **collecting, organizing** and **evaluating data**; **making deduction** and reaching at **conclusion** to determine whether they fit the **formulated hypothesis**.

- Research is simply the process of finding solutions to a problem **after through study and analysis of the situational factors.**
- It is gathering information needed to answer a question, and thereby help in solving a problem.
- Research can also be defined as **a scientific and systematic search** for gaining information and knowledge on a specific topic or issue.
- It is also known as the **art of scientific investigation.**

Business Research meaning

It is the application of the scientific method in searching for the truth about business phenomena.

Include defining business opportunities and problems, generating and evaluating alternative courses of action, and monitoring employee and organizational performance.

It is an organized, systematic, data-based critical, objective, scientific inquiry or investigation into a specific problem, undertaken with the purpose of finding answers or solutions to it.

it is mostly conducted to know where the problem areas exist in the organization, identify as clearly and specifically as possible the problems that need to be studied and resolved, and gather information, analyse the data, and determine the factors that are associated with the problem and solve it.

Motivation of Doing Research

- Research is **not a trouble-free** duty rather it is time consuming, **tiresome** and expensive undertaking.
- Despite these all difficulties many people especially in academic institute would like to carry out research at least **once** in their life.

Question

What are the **possible motives** of individuals to undertake research work?

• The possible motives for doing research can be either one or a combination of the following:

- Desire to **get degree with** its consequential benefit.
- Desire to **get respect and promotion** (to own respect in society): “*Publish or perish*” is a phrase commonly used in experienced and well-established academic institutes.

Motives Cont'd...

- Desire to face **a challenge** in solving the unsolved problem.
- Desire to get **intellectual joy** of doing some creative work.
- **Directive** of government.
- Required **criteria** for some jobs.

Importance of Research

- All progress is born of **inquiry**. Doubt is often better than overconfidence, for it leads to inquiry, and inquiry leads to **invention**.
- All development and progress in the society is an outcome of a research work.
 - Research provides base for the policies formation of the Government
 - Decision-making may not be a part of research, but research certainly facilitates the decisions of the policy maker.
 - Research has special importance in relation to solving various problems of business and industries.
 - research may means of careerism or a way to attain a high position in the social structure.

Managerial Value of Research

- The prime managerial value of business research is that it reduces uncertainty by providing information that improves the decision-making process.
- The decision making process associated with the development and implementation of a strategy involves;
 - Identifying problems or opportunities
 - Diagnosis and assessment problems or opportunities
 - Selecting and implementing a course of action
 - Evaluating the course of action

Qualities of Good Research

- **Good research is systematic:** It means that research is structured with specified **steps** to be taken in a specified sequence in accordance with the well defined **set of rules**.
- **Good research is logical/rational:** This implies that research is guided by the rules of **logical reasoning**.
- **Good research is empirical:** It implies that research is related basically to one or more aspects of a real situation.
- **Good research is replicable:** This characteristic allows research results to be **verified** by replicating

Types of Research

I. On the basis of the outcome of the research:

a) Fundamental Research:

- also called **academic or basic or pure** research.
 - aimed at investigating or search for **new principles and laws**
 - concerned with generalization and formulation of **a theory**
 - **Verifies** the old established theories, principles and laws
- E.g. **Darwin Theory of Evolution**
 Newton's Law of Motion
 Einstein Theory of Relativity

b) Applied research

- Aimed at finding a solution for an **immediate problem facing** a society, a group or industry.
- While **pure research** discovers **principles and laws**, **applied research** discovers **their application** in solving some social, economical or any **other problems**.

E.g.

- Investigation into the **ways** of improving **employee motivation**
- Study into the ways of improving the **levels of customer retention**

Types of Applied researches

a) Social impact Analysis:

- Major purpose of social impact assessment is to estimate the likely consequences of a planned change.
- It is the Process of **identifying** and **managing** the social impact of Industrial project, **policies, programs, plans**
- can be used for **planning** and making choices among **alternative policies**

- **Potential areas assessed in social impact analysis.**
 - Community service (School enrolment, speed of policy respond)
 - Social conditions (crime rate etc)
 - Economic impact (change in income level)
 - Demographic consequences (change in the mix of old and young people, population movement in or out of the area)
 - Environment impact
 - Health outcomes

b) Evaluation research:

- addresses the question, “**did it work?**”
- measures the **effectiveness of a program**, policy, or way of doing something.
- There are two types of evaluation research; **Formative** and **Summative**.

e.g. Your assessment schemes

II. On the basis of the purpose

a) Exploratory research (Pilot Survey)

- Also called **preliminary research**
- Aimed at discovering, identifying and formulating a research problem and hypothesis.
- Conducted when there are few or no studies previously done on a given problem.

b) Descriptive Research

- The main purpose of such research is description of the state of nature or affairs, as it exists at present
- we often use the term **ex-post facto research** for descriptive research
- The researcher **has no control over the variables**; he/she can only report **what has happened** or **what is happening**

- **Examples**

- Assessing the absentee rate in a particular office
- Describing the qualification of employees in a given organization
- Studying the preferences of customers at selective supermarkets
- Evaluating the academic performance (Achievement) of Accounting and Finance students in case of Jimma University

c) Analytic research

- Analytical research, goes beyond **simple description** of the state of nature.
- When a researcher encounters an issue that is already known and have a description of it, you may begin to ask **“why” things are the way they are.**
- Analytical research uses facts or information already available, and analyzes them to make a critical evaluation of the material

- Not only describe the characteristics, but also it analyses and explains **why and how** it happened or is happening
- Some of the types of analytical research are **case study** and **tracer (follow up) study**.
- **Examples**
- Investigating the **possible ways of reducing** the customers complaints toward the product price
- Studying the **cause of students** low achievement and looking for **possible remedies**
- **etc,.**

d) Predictive research

- Such research goes **beyond explaining why and how** things happened.
- It **predicts (forecast)** the probability of happening similar situation in other places or in the future
- It tries to explain about the probability of happening **similar thing in the future.**
- **Example**
- **How could inflation affects the country's economy?**
- **Do Covid-19 has impact on the livelihood of Coming generation?/ Continue to exist in the long run?**

III. On the basis of the type of data used:

- a) Qualitative research:
- b) Quantitative research:

IV. On the basis of the environment in which the research is carried out

- a) Field research
- b) Laboratory research
- c) Simulation research (create a model that displays the behavior/operation of the actual process/system)

V. On the basis of the time required to complete the research

- a) One -time research
- b) Longitudinal research

	Qualitative	Quantitative
Conceptual	Concerned with understanding human behaviour from the informant's perspective Assumes a dynamic and negotiated reality	Concerned with discovering facts about social phenomena Assumes a fixed and measurable reality
Methodological	Data are collected through participant observation and interviews Data are analysed by themes from descriptions by informants Data are reported in the language of the informant	Data are collected through measuring things Data are analysed through numerical comparisons and statistical inferences Data are reported through statistical analyses
<i>Source: Adapted from Minichiello et al. (1990, p. 5)</i>		

Research and Scientific method

- The researcher is not interested in particular results; rather he is interested in the repeatability and validity of results and their application to more complicated and general solutions.
- Research methodology and technique are mostly different from one science to another or can be different from research to research. **The philosophy common to all research methodology and technique is called Scientific methods.**

Steps in scientific method

1. **Observation:** the first or initial step in the scientific method is a minute and careful observation of the subject matter of research.
2. **Recording:** is a careful recording of all information (data) obtained in the first step.
3. **Classification:** implies systematically arranging and organizing the recorded data on the **logical basis** so that they will become viable for the analysis and generalization
4. **Generalization:** is an **preparing** of general laws and principles on the basis of the pattern exhibited by the classified material
5. **Verification:** the general principle and laws must be also verified

Criteria of scientific research

- **Purpose clearly defined:** The purpose of the research should be clearly defined and common concepts be used.
- **Research process detailed:** The research procedure used should be **described in sufficient detail** to permit another researcher to repeat the research for further advancement, keeping the continuity of what has already been attained.
- **Research design thoroughly planned:** The **procedural design** of the research should be **carefully planned** to yield results that are as **objective** as possible.

- **Limitations frankly revealed:** The researcher should report with complete frankness, flaws in procedural design and estimate their effects upon the findings.
- **High ethical standards applied:** A research design that includes safeguards against causing mental or physical harm to participants and makes data integrity a first priority should be highly valued.
- **Adequate analysis for decision maker's needs:** The analysis of data should be sufficiently adequate to reveal its significance and the methods of analysis used should be appropriate.

- **Findings presented unambiguously:** The validity and reliability of the data should be checked carefully.
- **Conclusions justified:** Conclusions should be confined to those justified by the data of the research and limited to those for which the data provide an adequate basis.
- **Researcher's experience reflected:** Greater confidence in research is warranted if the researcher is experienced, has a good reputation in research and is a person of integrity

Methods versus Methodology

Research methods:

- *Research methods* may be understood as all those methods/techniques that are used for conduction of research.
- *Research methods or techniques, thus, refer to the methods the researchers use in performing research operations.*

- Keeping this in view, research methods can be put into the following three groups:
 1. approaches or techniques of data collection
 2. those statistical techniques used to establish relationship between variables
 3. those techniques involved in evaluating the accuracy of the result obtained

Research methodology:

- **Research methodology** is a way to systematically solve the research problem.
- It may be understood as a science of studying how research is done scientifically.
- ***Thus, when we talk of research methodology we not only talk of the research methods but also:***
 - ***consider the logic behind the methods we use in the context of our research study***
 - ***explain why we are using a particular method or technique and***
 - ***why we are not using others***

Research Processes

- Research processes consist of a series of actions or steps, which are necessary to successfully carry out research activities.
- consist of a number of closely related activities.
- These activities (steps) can overlap continuously rather than following a strictly prescribed sequence.
- The first step can determine the nature of the last step to be undertaken.
- These steps are not separate and distinct.
- They do not necessarily follow each other in any specific order

- The following order of steps provides a useful procedural guideline regarding the research processes.
 - Identification and Formulating the research problem
 - Extensive literature survey
 - Developing working hypothesis
 - Preparing the research design
 - Determine sample design
 - Collecting data
 - Execution of the project
 - Analysis of the data
 - Hypothesis testing
 - Generalization and interpretation
 - Reporting the result

CHAPTER TWO

RESEARCH PROBLEM AND HYPOTHESIS FORMULATION

WHAT IS A RESEARCH PROBLEM?

- A research problem, in general, refers to **some difficulty which a researcher experiences** in the context of either a theoretical or practical situation and wants to **obtain a solution for the same.**
- A research problem is a **specific issue, difficulty, contradiction,** or **gap in knowledge** that you will aim to address in your research.

• We say a research problem **exist**, if the following conditions are met;

- There will be **an individual (or group or organization)**, to whom the problem can be attributed.
- There must be at **least two courses** of action; that can be taken to solve the problem.
- There must be at least **two outcomes (Objectives)**, of the courses of action of which one should be preferred to the other.
- The courses of action available must provide some chance of obtaining the objective, but they cannot provide the same chance, otherwise the choice would not matter.

We can state the components of a research problem as under:

1. There must be an individual or a group which has some difficulty or the problem.
2. There must be some objective(s) to be attained at. If one wants nothing, one cannot have a problem.
3. There must be alternative means (or the courses of action) for obtaining the objective(s) one wishes to attain.
4. There must remain some doubt in the mind of a researcher with regard to the selection of alternatives.
5. There must be some environment(s) to which the difficulty pertains.

NECESSITY OF DEFINING THE PROBLEM

- *A problem clearly stated is a problem half solved.*
- A proper definition of research problem will enable the researcher to be on the track whereas an ill-defined problem may create hurdles.
- A properly defined research problem will give answers to the following questions:
 - What data are to be collected?
 - What characteristics of data are relevant and need to be studied?
 - What relations are to be explored?
 - What techniques are to be used for the purpose?

TECHNIQUES INVOLVED IN DEFINING A PROBLEM

1. Statement of problem in a general way:

- the problem should be stated in a broad general way keeping with some **practical**, **scientific** and **intellectual interest**.
- For that purpose the researcher must immerse himself completely in the subject matter, why he wishes to pose a problem.
- In social science it is advisable to do some field observation and/ or preliminary survey (pilot survey).

2. Understanding the nature of the problem:

- The next step is to understand clearly the nature and the origin of the problem.

The best way to understand the nature of the problem is:

- discuss with those who first raised the problem
- discuss it with those who have a good knowledge of the problem concerned or similar other problem

3. Survey the available literature:

- All available literature concerning the problem must be studied and examined before defining research problem.

This means the researcher must be familiar with:

- The relevant theory in the area.
- Reports and records and other literature in the concerned area
- Review research works undertaken on related problem.

• Generally, survey literature will enable researcher to know

- If there are certain gap in the theory
- Whether the existing theory applicable to the problem and consistent with each other.
- Whether the findings of the research do or do not follow a pattern consistent with the theoretical expectation.
- Study on a related problem is also useful for indicating the type of difficulty that may be encountered in the present study.

4. *Developing ideas through discussion:*

- The researcher should discuss his problem with colleagues and others who have enough experience in the same area. Such practice is called “**experience survey**”.
- Peoples with rich experience are in a position to show the researcher different aspects of his proposed study and their advice and comments are usually of high values.

5. Rephrasing the research problem :

- The problem that has been stated in broad and general way should be reformulated to working proposition.
- The researcher should narrow and break down the problem into its components variables and relationship.
- Problem should be expressed as:
 - a relationship between two or more variable
 - the problem should be stated either in question form or hypothesis form

- Question form is appropriate mostly when the research is **descriptive** in nature.
- The formulated problem should be free from ambiguity and the relationship among variables should be clearly expressed .
- **Examples**
 - *Does a relationship exist between income of university students and score on their exams?*
 - *Is there a relationship between employees' age and their productivity?*

The following points must be considered while redefining the research problem:

- Technical terms and words should be clearly defined.
- Basic assumptions or postulates (if any) relating to the research problem should be clearly defined.
- The criteria for the selection of the problem should be provided.
- The scope of the investigation must be mentioned explicitly .

EVALUATION OF THE RESEARCH PROBLEM

- Research problem should be evaluated in terms of the following criteria
- 1. Is the problem researchable?**
 - Some problems cannot be effectively solved through the process of research.
 - Therefore, the problem must be stated in workable research question that can be answered empirically.
 - 2. Is the problem new?**
 - As much as possible, the research problem needs to be new.
 - The researcher should not target to the problem that had already been thoroughly investigated by other researchers.

- To be **safe** from such duplication, the researcher has to go through the record of **previous** studies in a given field.
- However, there are times where by a problem that has been investigated in the past could be **worthy of study**.
- A researcher may repeat a study when he wants to **verify its conclusion** or to extend the validity of its findings in situation **entirely different** from the previous one.

3. Is the problem significant?

The question of significance of the problem usually relates to:

- what a researcher hopes to accomplish in a particular study?
- What is his purpose in undertaking to solve the particular problem he has chosen?
- What new knowledge does he hopes to add to the sum total of what is known/ Research **gap**?
- When these all questions are answered clearly by the researcher, the problem should be considered for **investigation**.

4. Is the problem feasible/achievable?

- The feasibility of the research problem should be examined from the point of view of the researcher's personal aspects as stated here under:
 - Researcher Competence
 - Interest and enthusiasm/eagerness
 - Financial consideration
 - Time requirement
 - Administrative consideration

CHARACTERISTICS OF A GOOD RESEARCH PROBLEM

- ❖ The problem can be stated clearly and concisely.
- ❖ The problem generates research questions.
- ❖ It is grounded in theory.
- ❖ It relates to one or more academic fields of study.
- ❖ It has a base in the research literature.
- ❖ It has potential significance/importance.
- ❖ It is attainable within the time frame, budget.
- ❖ Sufficient data are available or can be obtained.
- ❖ The researcher's methodological strengths can be applied to the problem.
- ❖ The problem is new; it is not already answered sufficiently.

Sources of Research problem

- **Professional experience**
- **Extensive study of available literature**
- **Inference from theory**
- **Technological and social changes**
- **Consultation/ Discussion**
- **Previous Studies**

The following points may be considered in selecting a research problem.

- A subject that is **overdone** should not be normally chosen, for it will be a difficult task to throw any new light to such a case. So, The topic selected should be **original** or at least less explored.
- It should be big enough to be researchable and small enough to be handled. **Too narrow or too vague/unclear** problems should be avoided.
- The subject selected for research should be **familiar** and **feasible**.
- It should be interesting to the researcher and should fit into his aptitude.

Cont....

- It should be of significance and socially relevant and useful.
- The selection of a problem must be preceded by a preliminary study (if the research relatively new).
- The importance of the subject, the qualification and the training of researcher, the cost, the time factor are the few points that must be considered in selecting a problem.

HYPOTHESIS FORMULATION

- **HYPOTHESIS Defined.....**
- It is the composition of two words, “**Hypo**” and “**Thesis**”.
Hypo means “**under or below**” and thesis means “**a reasoned theory or a rational viewpoint**”.
- Thus hypothesis means a theory which is **not fully reasoned**.
- Hypotheses are a set of suggested tentative solution of a research problem, which **can be or may not be** a real solution.
- “It is a **tentative/uncertain** prediction about the nature of the relationship between two or more variables.”
- it states the researcher’s expectations concerning the relationship between the variables in the research problem

- a hypothesis is the **most specific** statement of the problem.
- It states what the researcher thinks the **outcome** of the study will be.
- The researcher does not then set out to “**prove**” his or her hypothesis rather collects data that **either support** the hypothesis or **do not support it**;
- Research hypothesis differs from research question in that, hypothesis both indicate the question in testable form and predict the nature of the answer.
- The task of the researcher in this case will be to establish and test such hypothesis.

- A hypothesis is future-oriented.
- The hypothesis is formulated following the review of related literature and **prior to the execution** of the study.
- It logically follows the review since it is based on the implications of previous research.
- Every aspect of the research is affected by the hypothesis (if any), including subjects (the sample), measuring instruments, design, procedures, data-analysis techniques, and conclusions.
- The **variables** must be clearly specified and measurable by some known techniques

Source of research Hypotheses

- **Originates in the science itself or in the scientist's Experience**
- **Resulted after deep analysis and Comparisons of phenomenal**
- **Resulted from findings of other studies**
- **Originates from Theory**
- **From imagination of the researcher etc...,**

Importance of Hypothesis

A well-grounded hypothesis provides the following advantages

- Represents specific objective, which determine the nature of the data needed to test the proposition
- Offer basis for selecting the sample, the research procedure, and the statistical analysis needed.
- Keeps the study restricted in scope thereby preventing it from becoming too broad
- Sets a framework for reporting the conclusion of the study

Characteristics of Good hypotheses

Good hypotheses are;

- clearly and precisely formulated
- formulated in such way that, they can be **tested or verified**
- state explicitly the **expected** relationship between variables
- **limited** in scope.
- **consistent** with the known facts.
- stated in **simple terms**.

TYPES: Null and Alternative Hypothesis

- The null hypothesis (H_0) represents a theory that has been put forward, either because it is **believed to be true** or because it is to be used as **a basis for argument**, but has not been proved.
- states a definitive, **exact** relationship between two variables.

- Expressed as **no (significant)** relationship between two variables or no (significant) difference between two group.

Example

- There is **no** relationship between student achievement and their income.
- The relationship between profitability of SME and their employees level of education is **zero**.
- Political participation is **not** the result of education.

- The **alternative hypothesis (H_a)** is a statement of what a hypothesis test is set up to establish.
- **Opposite** of Null Hypothesis.
- Only reached if H_0 (null hypothesis) is rejected.
- Frequently “alternative” is actual desired conclusion of the researcher.
- Example
- Formulate ‘ H_a ’ using examples provided under ‘ H_0 ’

Steps in Hypotheses Testing

- State the null and the alternate hypotheses
- Choose the appropriate statistical tests based on data collected (parametric or non parametric)
- Determine the level of significance desired
- See if the output results from computer analysis indicate that the significance level is met.
- Make decision (Either to reject or accept null hypotheses)
- Example



CHAPTER THREE

Research Proposal and Guide to Prepare a Proposal

What is a Research Proposal?

- The research proposal is the detailed plan of study.
- The term "research proposal" indicates that a specific course of action will be followed.
- The intent of the written research proposal is to present a focused and scholarly presentation of a research problem and plan.
- The objective in writing a proposal is to describe what you will do, why it should be done, how you will do it and what you expect will result.

- A research proposal can be defined as a **written document** requesting both authorization and funds to undertake a specific research project.
- It always includes a statement explaining the purpose of the study or a definition of the problem.
- It systematically outlines the particular research methodology and details the process that will be utilized at each stage of the research process

Function of Research Proposal

i. Proposal as a means of communication:

- Serves to communicate the investigator's research plan to those who-give consultations and/or disburse fund.

ii. Research proposal as plan:

- Helps the researcher to organize his idea in a systematic manner and to look for strengths and flaws.
- A successful proposal sets out the plan in step-by-step detail.
- It provides an inventory of what must be done and which materials have to be collected as a preliminary step.

ii. Proposal also functions as a contract:

- A completed proposal approved for execution and signed by all members of the sponsoring committee, constitute a bond of agreement between the researcher and that committee.
- An approved grant proposal results in a contract between the investigator (and often the university) and funding sources.

General Format and components of Proposal

1. Title
2. Abstract
3. Introduction (background and justification), Objective of the study
4. Literature review
5. Material and methods (methodology)
6. Work plan (time budget)
7. Cost budget
8. References
9. Appendix

1. Title

- The title of research proposal should state the topic exactly in the smallest possible number of words.
- **All words in the title should be chosen with great care, and association with one another must be carefully managed.**
- A good title is defined as the **fewest possible words** that adequately describe the contents of the study.
- Titles should almost **never contain abbreviations.**
- The title page has no page number and it is not counted in any page numbering.

2. Abstract

- **Is summary**, which reflects the whole content of the proposal (most of the time **less than 300 words**).
- Abstract should be concise, informative and should provide brief information about the whole problem to be investigated.
- Do not put information in the abstract **that is not** in the main text of your research proposal.
- Do not put references, figures, or tables in the abstract.

The abstract of a proposal should contain the following points ;

- Title or topic of the research
- Statement of the problem and objective
- Methodology of investigation
- Expected result (tentative only if a researcher starts with a formulated hypothesis) .

3. Introduction

- This section in research proposal should give the general background of the study and the need for undertaking the study.
- It reviews the previous work to reveal what is known of the subject.
- You need to demonstrate that you know what you're talking about and that you have knowledge of the literature surrounding this topic.
- This is a statement of something sufficiently interesting to motivate your reader to read the rest of the proposal.

- You should then go on to explain why more work is necessary (your work, of course).

Some question to be addressed in this part may include:

- What are current and previous studies that have been made on the issues to be studied?
- What is the present gap in knowledge and current trend?
- What makes worth studying?
- Why has the problem not been solved yet?
- What do we intend to fill the gap or solve the problem?

4. *The Rationale*

- The Rationale provides some of the reasons for doing the study and its importance.
- **Citations from past research** may be relevant here; likewise recent political and economic events, previous efforts to solve the problem that did not succeed can be mentioned.

5. *Objectives*

- The objectives of a research project summarize what is to be achieved by the study.
- Objectives should be closely related to the statement of the problem.

Objectives should be

- simple (not complex),
- specific (not vague),
- stated in advance (not after the research is done), and
- Stated using “action verbs” that are specific enough to be measured.

6. *Literature review*

- Literature review means locating literature in a variety of sources, reading it carefully and thoroughly organizing it into themes along with the line of investigation.

Objective of the literature review:

- To establish a familiarity with a body of knowledge and establishes credibility
- To show the path of prior research and how a current project is linked to it
- To integrate and summarize what is known in the area
- To learn from others and stimulate new ideas.

7. Material and Methods

- The methodologies are the core and the largest part of the scientific research proposal.
- The methodologies are largely descriptive.
- This section is an essential prerequisite for validating the results and hence acceptability of the proposal.
- Should give full details to show how the research activity is going to be carried out.

- The proposal should describe in detail the general research plan. (May **not necessarily** be true for all types of research)
 - Description of study area
 - Description of study participants (Target Population)
 - Determination of sample size
 - Description of selection process (sampling method)
 - Methods of data collection
 - Operational definition
 - Method of data analysis, Presentation and Interpretation

8. *Budget of Time*

- A timetable explaining how the researcher expects to carry out his project and when each of the important phases will be completed is helpful to both the researcher and the reviewer.
- It is a plan in terms of numbers of weeks or months and expected completion dates.
- Commonly researcher presents it in a table form.

9. Reference (Bibliography/Literature cited)

- All works cited in the proposal are listed either alphabetically or numerically at the end of the document usually under the heading of “**References or Bibliography**”
- There are several ways of making citation and recording references. The two main systems of referring are:
 - A) **Harvard System**: This system uses author’s name, date, and page reference.
 - B) **Vancouver System**: uses in-text number instead of an author, date, and page reference.

- *Harvard System*

Since this system is used in business research it is presented in some detail below.

- Each reference to a periodical publication must include, in order, the name(s) of the author(s), the year of publication, the full title of the article, the publication in which it appears, and the volume and inclusive page number.
- Reference to a book must give the author or authors' name(s), the year, the title, the edition, if other than the first; the publisher, the city of publication, and the number of the volume (if two or more).

10. Appendix (Annex)

- It is not mandatory to have this section.
- If the researcher thinks that having this section will increase the quality of the research proposal he is free to do so.
- Information types provided in this section are those additional details, which are difficult to accommodate within the standard headings.

Stuffs to be included in appendix:

- Detail Questionnaires and interview schedules.
- Detailed experimental design (in experimental research)
- Detailed statistical procedures.
- Samples of research materials (chemical and some biological materials).
- Survey and other geographic maps.



CHAPTER FOUR

RESEARCH DESIGN

Meaning of Research Design

- A researcher after defining and properly formulating a research problem as a next step, he has to prepare a research design or plan of the research work.
- ***Research design*** is a plan of collecting and analyzing data in an economic, efficient and relevant manner.
- It is a plan of organizing framework for doing the study and collecting the necessary data.
- It constitutes the blue print for collection, measuring and analysis of data.

- The design decision should be oriented in answering the following question.
 - What is the study about?
 - Why is the study being made?
 - Where will be the study being carried out? What types of data are required?
 - Where can the required data be found and what techniques of data collection will be used?
 - What period of time will the study include?
 - What will be the sample design?
 - How will be the data analyzed?
 - In what style will the report be prepared?

Need for research Design

- It facilitates the smooth sailing of the research operation
- It makes research project as efficient as possible and help to yield maximum information with minimum expenditure, time and effort.
- It helps the researcher to organize his ideas in a form where by it will be possible for him to look for flaws and inadequacies
- Design will be given to others for their comment and critical evaluation.

Important concepts relevant to research design

- ***Dependent variable:*** If one variable depends upon or a consequence of the other variable is called a dependent variable. Is a variable that is to be predicted or explained.
- ***Independent variable:*** is a variable that is expected to influence the dependent variable.
- ***Extraneous variable:*** Independent variables that are not related to the purpose of a study, but may affect the dependent variable are termed as extraneous variable.

- ***Control:*** A good research design has to minimize the influences of extraneous variable. To do so the researcher uses control as a remedy to minimize the effects of extraneous variable.
- ***Confounded relationship:*** When dependent variable is not free from the influence of extraneous variable, then the relationship between dependent and independent variables is said to be confounded an extraneous variable.
- ***Experimental and control groups:*** In experimental research when a group is exposed to usual condition is called control group, but when a group is exposed to special condition is an experimental group.

- ***Treatments:*** The different condition under which experimental and controlled groups are put are referred to us treatment.
- ***Experiment:*** The process of examining the truth of a statistical hypothesis, relating to some problem, is known us an experiment.
- ***Experimental unit:*** the pre-determined plots (or blocks or group) where different treatments are used are known experimental units.

Forms of research Design

Research design for non-experimental research

A. Design for Exploratory research study:

- Exploratory research is a research aimed at identifying and formulating a research problem or developing working hypothesis.
- The main purpose of such study is only formulating a problem for further, precise investigation or of developing the working hypothesis from an operational point of view.

- The design for such study is characterized by a great deal of *flexibility*.
- Since no clear hypothesis has been developed about the problem, the following forms of research design can be used.

i. The survey of concerning literature:

- This is the most simple and fruitful method of formulating precisely the research problem or developing hypothesis.
- Hypothesis stated by earlier works may be reviewed and their usefulness be evaluated as a basis for further research.

ii. Experience survey:

- This implies the survey of people who have practical experience with the problem to be studied.
- The objective of such survey is to obtain insight into the relationship between variables and new ideas relating to the problem.
- For such purpose people who are competent and can contribute new ideas are carefully selected as respondent and the investigator then interview them

iii. Analysis of insight (Examining analogous situation):

- Is also important method for suggesting hypothesis for research.
- It is particularly suitable in areas where there is a little experience to serve as a guide.
- This method consists of intensive study of selected instances of the phenomenon.

B) Research design in case of descriptive and diagnostic research studies:

- Descriptive research studies are those studies which are concerned with describing the characteristics of a particular individual, or of a group, whereas diagnostic research studies determine the frequency with which something occurs or its association with something else.
- Descriptive research in contrast to exploratory research is marked by the prior formulation of specific research problem (question).

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- Investigator already knows a substantial amount about the research problem, may be as a result of exploratory study, before the project is initiated.
 - Thus, the researcher should be able to define clearly, what he wants to measure and setup appropriate and specific means for measuring it.
 - Since the purpose is to obtain complete and adequate information, the procedure to be used must be carefully planned.

- Thus the design in such study must be rigid and must focus on the following:

a) Formulating the objective of the study:

- The first step in descriptive study is to specify the objective with sufficient precision to ensure that the data collected are relevant.

b) Design the method of data collection:

- First of all the researcher has to determine the relevant and type of data needed to the study.
- After defining the relevant information needed for the designed study, method of data and technique for collecting them must be devised.

c) Select the sample of the study (sampling):

- There are different sampling techniques among which the appropriate sampling technique would be selected.
- The sample expected to be representative and yielding accurate information with a minimum amount research effort.

d) Collect the data relevant for the study:

- It is necessary to supervise closely the staff or field workers as they collect and record information.
- As data are collected, they should be examined for completeness, comprehensibility, consistency and reliability.

e) Processing and analysis of the data collected:

- Data processing design includes steps like coding the interview replies, observation, tabulating the data; and performing several statistical computations.
- To the extent possible, the processing and analyzing procedure should be planned in detail before actual work started.

f) Reporting the findings:

- This is the task of communicating the findings to others and the researcher needs to be well planned so that all things relating the research study may be well presented in simple and effective style.

Research design for experimental research

- Experimental studies are those in which the researcher can control and manipulate at least one of the independent variable and test the hypothesis of causal relationship between variables.

Basic principles of experimental design

1. The Principle of Replication:

- The experiment should be repeated more than once
- Each treatment is applied in many experimental units instead of once
- By doing so the statistical accuracy of the experiment is increased
- Replication is introduced in order to increase the precision of the study
- The entire experiment can even be repeated several times for better results

2. The Principle of Randomization:

- This principle provides protection, when we conduct an experiment, against the effect of extraneous factors by randomization

3. The Principle of Local Control:

- The extraneous factor, the known source of variability, is made to vary deliberately over as wide a range as necessary and this needs to be done in such a way that the variability it causes can be measured and hence eliminated from the experimental error.

- In other words, according to the principle of local control, we first divide the field into several homogeneous parts, known as blocks, and then each such block is divided into parts equal to the number of treatments. Then the treatments are randomly assigned to these parts of a block.

Sources of Information

- Any research is based on various types of information.
- The more information the researcher has about the phenomenon the better will be his investigation, and explanation about it. Without information no investigation will be made.

- Different authors have classified the sources of information differently. Some of the most important views are of the following:
- **According to P.V. Young:** The sources of information can be classified into:
 - a. **Documentary sources:** These include books, manuscripts diaries and letters. Documents first written out (not published yet) or not printed are also included here.
 - b. **Field sources:** The information given by individual, obtained from field experiment, survey and observation

- **According to W. A. Bagley**, the sources of information in the field of social science may be classified into:
 - a. **Primary source:** Include the actual information received from individuals directly concerning the problem of the study. It also includes observed social phenomenon and facts that may be discovered.
 - b. **Secondary sources of information:** These include all types of published and unpublished, public or private documents and other such types of information.

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- **According to G.A. Lundberg:** The sources of information may be classified as:
 - a. **Historic sources**, which includes sources representing the past incident, for example, documents, papers and stone inscriptions concerning past articles discovered through excavations.
 - b. **Field sources:** Include information received from concerned persons and also through the observation of their behavior.

- Generally, the sources of information are of two types, i.e., **primary sources and secondary sources of information.**

1. Primary sources of information: The information obtained from primary sources is often called as primary data.

- **Primary data:** Are those information gathered by researcher himself and are gathered for the first time, thus, happening to be original.
- Such data are known as primary because they are gathered by the researcher himself from the field of research directly and for first time.

2. Secondary sources of information: The information obtained from secondary sources is often called “secondary data”.

- **Secondary data:** are those, which have already been collected by some one and which, have already been passed through statistical process.
- The sources of secondary data are:
 - Personal document such as diaries, letters, photographs and so on.
 - Public documents such as books, manuscripts, records, census reports of survey by private and public institution; and various information published in newspapers and magazines.



Methods of Data Collection: Primary Data

Observation method of data collection

- Under this method the researcher should personally and directly observe the condition and incidence of his fields of study.
- The researcher would not ask anything from respondents.
- Direct observation is the most reliable method for gathering information related to the life style, status, conduct, behavior, language, custom and tradition and the like phenomenon.



The researcher using this method should decide priority:

- What should be observed?
- How the observation should be recorded?
- How the accuracy of observation can be ensured?

Participant Vs Non-participant observation

- A. Participant observation:** The researcher lives in the group or in the community as a member of it and participates in their life. He experiences what the member of the group experience.
- B. Non-participant observation:** The researcher does not participate in the group life but observe as an external spectator. Under such approach the presence of the researcher is unknown to the people.

Control Vs Uncontrolled Observation

- A. Uncontrolled observation:** Is the observation, which takes place in natural setting. Here no attempt is made to use precision instruments. The main aim of this type of observation is to get a spontaneous picture of life or person.
- B. Controlled observation:** Observation takes place according to definite pre-arranged plans, including experimental procedure. In this observation we use mechanical (or precision) instruments as aids to accuracy and standardization.

Structured Vs Unstructured

- A) Structured Observation:** It is structured and preplanned observation technique. This observation is characterized by a careful definition of units, the style of recording the observation information; standardize conditions of observation and the selection of pertinent observation.
- B) Unstructured observation:** The observation takes place without the characteristics mentioned above, i.e., without standardized format and plan. There is no specification of subject of observation, behavior to be observed and no standard format for recording the observed information.

Interview Method

- In an interview, the researcher meets people and discusses his social economic or some other problem with them to gathers facts.
- An interview is different from Schedules (interview questionnaires).
- Interview can be either personal **interview** or **interview** through telephone.

Personal Interview

- Such method requires interviewer-asking question in face-to-face contact to respondent.
- The interviewer has to be on the spot and has to meet people from whom the data are collected.
- This approach is suitable for intensive investigation.
- The personal interview can be of two type - **structured and unstructured**
- This approach is used in exploratory research where as the structured interview is used in descriptive research.

Telephone Interview

- This method of collecting information involves contacting respondents on telephone.

Prerequisites of interview

- Interviewer should be honest, sincere, hardworking, impartial and must possess the technical competence
- Interviewer has to create friendly atmosphere, trust and confidence
- Interviewer must ask question properly and completely
- The interviewer should not show surprise or disapproval of a respondent's answer

Basic principles of Interviewing

- Ask only one question at a time
- Repeat the question if necessary
- Listen carefully to the subjects answer
- Observe the subjects facial-expression, gesture and tone of the voice
- Allow the subject sufficient time to answer the question

Collection of data through Questionnaires

- Questionnaire is a list of structured questions, which will be presented, mailed or e-mailed to selected respondents to obtain reliable response from them.
- This method of data collection is used when the subject of study is very wide and direct observation is not possible.

Types of Questionnaires

Questionnaires can be of the following type:

- Interview Questionnaires (Schedules)
- Mail Questionnaires (Self administered questionnaires)
- Questionnaires through Internet (Through electronics media)

Mail Questionnaires

- This method of data collection is very popular especially for large study.
- The questionnaires are sent by post to the person concerned with a request to answer the question and return them back.
- Questionnaires are mailed to respondents who are expected to read and understand the question and write down the answer.

Guide to question construction

A) Forms of questions

- Open-ended questions
- Closed ended questions

B) Make items clear

- Make questionnaires item clear
- Avoid double –barreled questions
- Respondent must be competent to answer
- Avoid Negative items

C) General questionnaire format

i. Format for responses

- For closed-ended question
- For open-end question

ii. Ordering questions in questionnaires

iii. Opening question

iv. Questionnaire format

- Instruction
 - General instruction
 - Specific instruction
 - Introduction



V. Mail distribution and return: accompanied by a letter of explanation and a return envelope.

- *Monitoring the return*
- *Follow-up-mailing*
- *Response rate*

Schedule (Interview Questionnaires)

- Schedules are being filled in by enumerators (interviewers) who are specially appointed for the purpose
- The questionnaires are either delivered by hand to be answered by respondents or the trained interviewer will read the question to respondent and record the answer given.

General Rule for Interviewing

- Appearance and Demeanor (behavior or conduct)
- Familiarity with questionnaires
- Record responses exactly
- Probing for response
- Interviewer should be trained
- Interviewer should Practice both in class and field



Method of Secondary Data Collection

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- Researchers use secondary data for different reasons:
 - Secondary data may solve problem
 - Search costs are substantially lower than primary sources
 - It has also important supplementary use, these includes:
 - to define the problem and formulate hypothesis
 - to plan the collection of primary data
 - to define the population and select the sample in primary data collection

- Before using secondary data, we must see that they possess the following characters

1. Reliability of the data

- The reliability can be tested by finding answers to the following questions about the said data:
 - Who collect the data? (How is the reputation of the collector?)
 - What were the sources of the data?
 - Were they collected using proper method?
 - At what time were they collected?
 - Was there any bias of the collector?
 - What level of accuracy was desired?
 - Was it achieved?

2. Suitability

- The data that are suitable for one inquiry may not necessary be found suitable in another inquiry.
- The researcher must very carefully scrutinize the definition of various terms and units of collection used at the time of collecting the data from the primary sources originally.

3. Adequacy

- If the area and scope of the study is narrower than the area of present study the data are considered as inadequate.
- Because of the fact that information will be missed and the researcher needs additional sources to make it complete and usable.

- **Secondary data may be obtained either from:**
 - A. Internal (within the company or organization)
 - B. External sources (outside sources)

- A. Internal sources of secondary data:** All companies or organizations or any other organs keep record of the every course of conducting business or any other activities.
 - Orders received are filled
 - Cost are recorded
 - Sales' people reports are submitted., etc.

B. External sources of secondary data: The major sources of external data are:

- Government (federal, state, local)
- Trade association and trade press
- Periodicals
- Institution (research, academic)
- Commercial services and the like

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- The secondary sources of information can be classified into
 - A. *Personal Document*
 - B. *Public document*

A. Personal Document

1. Biography/Autobiography

- *Autobiographies* are life histories, life experience and incidents of prominent individuals written by them.
- *Biographies* are life history, experience and events of some prominent person written by some biographer.
- Some great political, business leaders, social reformers and eminent persons write their own autobiographies or some other people (biographer) write their biography.
- Such persons provide useful information concerning, social, economical, political, religious and cultural conditions and incidents of their time.

2. Diaries

- Many educated people have a habit of writing diaries concerning their social relationship and life incidents.
- Some write diaries to remember important incidents of their life.
- Some on the other hand write diaries to note things, which they cannot tell to anyone.
- Diary writing is a confidential and private performance.

3. Letters

- Letters are often used by researchers as evidence especially in studies like, history and sociology and psychology.
- Letters are the medium of expressions of feelings likes and dislikes attitudes, desires, emotion, ambitions and important incidents of life.
- they constitute one of the most reliable sources of information about the inter relationship, such as, friendship, love, marital affairs, family tension and divorce and the like.

4. Memories

- Some people are interested in writing down memories for their travels, participation in different activities.
- Such memories provide useful information in social research

B. Public and Official Document

- Public documents are information gathered from some governmental or non-governmental institutions.
- Some common public documents are of the following types:

1. Records:

- Most government and non-government institution preserves many types of records consisting of important information.
- For instant each police department keeps record of criminals. Company records like no employees, wage paid, growth rate sale etc.

2. Census report and statistics:

- A very important source of data collection is census report and the annual digest of statistics and statistical report of various state departments and national bodies.
- These reports are reported by experts and, therefore, their data are highly reliable.

3. Journals and magazines:

- These are common and widely used sources of information. Information obtained from these sources is very much reliable.

4. Newspapers:

- Published news (discussion on contemporary issues, reports of meetings and conferences) is very important sources.
- Economic newspaper usually report data like,
 - Daily stock price
 - Exchange rate
 - Interest rate
 - Daily commodity price
 - Price indices (report on quarterly basis)

5. Other documents:

- Under this category we can have information sources like, television, film, radio and public speech.



Sampling Design and Procedure

Some fundamental definitions

- ***Population***: Is the theoretically specified aggregation of survey elements from which the survey sample is actually selected.
- ***Sampling Frame***: Is the list of elements from which the sample is drawn
- ***Sample***: A subset or some part of a larger population
- ***Sample design***: Is a definite plan for obtaining a sample frame
- ***Sampling***: Is the process of using a small number or part of a larger population to make conclusion about the whole population.
- ***Element***: Is unit from which information is collected
- ***Statistic***: Is a characteristic of a sample
- ***Parameter***: Is a characteristic of a population.

Steps involved in sample planning (Sampling procedure)

A) Defining the population

- implies specifying the subject of the study
- Specification of a population involves identifying which elements (items) are included, as well as where and when.

B) Census Vs Sample

- Once the population has been defined, a choice must be made between census and sample

Advantages of census

- Reliability
- Detailed information

Limitation of census

- Expensiveness
- Excessive time and energy

Need for sampling

- The use of sample has the objective of estimating; testing and making inference about a population on the basis of information taken from the sample
- Sampling can save time and money
- Sampling may enable more accurate measurement, because sample study is generally conducted by trained and experienced investigator
- Sampling remains the only way when population contains infinitely many members

- It usually enables to estimate the sampling error
- If the choice of sample units is made with due care and the population is homogeneous, the conclusion of the sample survey can have almost the same reliability as those of census survey
- Sampling technique also enables researchers to obtain detailed study
- Sampling technique also enables researchers to obtain detailed study, as the number of sample units is fairly small these can be studied intensively and elaborately

Limitations of sampling technique

- Less accuracy
- Misleading conclusion
- Need for specialized knowledge
- If it is done by average researcher the selection is liable to error

Sampling technique is used under the following conditions

- When the number of units is very large
- When at most accuracy is not required
- Infinite population
- When census is impossible
- Homogeneity

Essentials of an ideal sample

- An ideal sample should fulfill the following four basic characteristics
 - Representative-ness
 - Independence
 - Adequacy
 - Homogeneity

C) Sample design

- Operationally, sample design is the heart of sample planning
- Sample design should answer the following:
 - What type of sample to use?
 - What is the appropriate sample unit?
 - What frame (list of sampling unit) is available for the population?
 - How are refusals and non-response to be handled?

D) Sample size determination

There are a number of sample size determination methods

- *Personal judgments*
- *Budgetary approach*
- *Traditional inferences*: we need to have information about the estimated variance of the population, the magnitude of acceptable error and the confidence interval.
- *Bayesian Statistics*: This is the selection of the sample size, which maximizes the difference between the expected value of information (EVI) and cost of sampling. That is, marginal cost of information (MCI) should be equal to Marginal value of information (MVI).

E) Cost of Sampling:

The sample plan must take into account the estimated cost of sampling.

F) Execution of sampling process:

Sampling Techniques

- Sampling techniques are basically of two types namely, non-probability sampling and probability sampling.

Non-probability

- Non-probability sampling technique does not give equal chance that each element of the population will be included in the sample.
- Non-probability sampling can be adequate if the researcher has no desire to generalize his findings beyond the sample

1. Quota sampling

- Under this sampling approach, the interviewers are simply given quotas to be full-filled from the different strata.
- The researcher must makes sure that the sample is as representative and generalize-able as possible.

2. Judgment (Purposive or deliberate) sampling

- In this approach the investigator has complete freedom in choosing his sample according to his wishes and desire.
- The experienced individual (researcher) select the sample based upon his judgment about some appropriate characteristics required from the sample members.
- The researcher selects a sample to serve a specific purpose, even if this makes a sample less than fully representative.

3. Snowball Sampling

- It is also known as **Multiplicity sampling** or **Multi-stage Sampling**.
- The term snowball comes from the analogy of the snowball, beginning small but becomes bigger and bigger as it rolls downhill.
- The major purpose of snowball sampling is to estimate characteristics that are rare in the total population.
- First initial respondents are selected randomly but additional respondent are then obtained from referrals or by other information provided by the initial respondent.

4. Convenience Sampling

- This is a "hit or miss" procedure of study.
- No planned effort is made to collect information
- This sampling technique is not scientific and has no value as a research technique.
- In general, the availability and willingness to respond are the major factors in selecting the respondents.

Probability Sampling

- All probability samples are based on chance selection procedures.
- Chance selection eliminates the bias inherent in the non-probability sampling procedure, because this process is random.
- Probability sampling is the most preferred type of sampling because of the following characteristics
 - sample units are not selected based on the desecration of the researcher
 - Each unit of the population has some known probability of entering the sample

1. Simple Random Sampling

- It is the basic sampling method in every statistical computation.
- Each element in the population has an equal chance of being included in the sample.
- It is drawn by a random procedure from a sample frame

2. *Systematic Sampling*

- Systematic sampling involves only a slight difference from simple random sampling.
- The mechanics of taking a systematic sample are rather simple.
- If the population contains N ordered elements, and sample size of n is required or desired to select, then we find the ratio of these two numbers, i.e., N/n to obtain the sampling interval.
- Systematic sampling may increase representative-ness when items are ordered with regard to the characteristics of interest

3. Stratified Sampling

- If population from which the sample to be drawn does not constitute a homogeneous group, stratified sampling technique is used in order to obtain a representative sample.
- Under this technique, the population is divided into various classes or sub-population, which is individually more homogeneous than the total population.
- The different sub-populations are called strata.
- Then certain items (elements) are selected from the classes by the random sampling technique.

- The following questions should be considered in the context of stratified sampling
 - *How to form strata?*
 - *How should items (elements) be selected from each stratum?*
 - *How many items to be selected from each stratum (sample size)?*

4. Cluster sampling

- In cluster sampling the primary sampling unit is no more the individual elements in the population rather it is say manufacturing unit, city or block of city, etc.
- After randomly selecting the primary sample unit (city, part of city), we survey or interview all families or elements in that selected primary sample unit.
- The area sample is the commonly used type of cluster sampling.
- But it is less precise than random sampling.
- Cluster sampling is used only because of the economic advantage it possesses.

5. Multi-stage sampling

- Items are selected in different stage at random.
- Multi stage sampling is a further improvement over cluster sampling.



Chapter Eight

Data Analysis

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- The goal of any research is to provide information out of raw data
 - The raw data after collection has to be processed and analyzed in line with the outline (plan) laid down for the purpose at the time of developing the research plan.

Data Processing

- Data processing implies *editing, coding and classification* of collected data so that they are amenable to analysis

Editing:

- Is a process of examining the collected raw data to detect errors and omission (extreme values) and to correct those when possible
- It involves a careful scrutiny of completed questionnaires or schedules
- It is done to assure that the data are accurate, consistent with other data gathered, uniformly entered, and as complete as possible
- Editing can be either field editing or central editing

Coding:

- Refers to the process of assigning numerical or other symbols to answers so that responses can be put into a limited number of categories or classes.
- Coding is used when the researcher uses computer to analyze the data otherwise it can be avoided

Classification:

- Most research studies result in a large volume of raw data, which must be reduced into homogeneous group.
- This is to classify the raw data or arranging data in-groups or classes on the basis of common characteristics
 - *Classification according to attributes*
 - *Classification according to class interval*

Data Analysis

- Data analysis is further transformation of the processed data to look for patterns and relations among data groups
- By analysis we mean the computation of certain indices or measures along with searching for patterns or relationship that exist among the data groups
- Analysis particularly in case of survey or experimental data involves estimating the values of unknown parameters of the population and testing of hypothesis for drawing inferences.
- Analysis can be categorized as
 - Descriptive Analysis
 - Inferential (Statistical) Analysis

Descriptive Analysis

- Descriptive analysis refers to the transformation of raw data into a form that will make them easy to understand and interpret
- Descriptive response or observation is typically the first form of analysis
- The calculation of averages, frequency distribution, and percentage distribution is the most common form of summarizing data.

- The most common forms of describing the processed data are:
 - Tabulation
 - Percentage
 - Measurements of central tendency
 - Measurements of dispersion
 - Measurement of asymmetry
 - Data transformation and index number

Inferential Analysis

- Most researcher wishes to go beyond the simple tabulation of frequency distribution and calculation of averages and / or dispersion
- When the population is consisting of more than one variable it is possible to measure the relationship between them
- If we have data on two variables we said to have a bivariate variable, if the data is more than two variables then the population is known as multivariate population

- In case of bivariate or multivariate population, we often wish to know the relationship between the two or more variables from the data obtained.
- There are several methods of determining the relationship between variables
- Two questions should be answered to determine the relationship between variables.
- Is there association or correlation between the two or more variables? If yes, then up to what degree?
- This will be answered by the use of correlation technique. Correlation technique can be different

- In case of bivariate population; correlation can be found using
 - Cross tabulation
 - Karl Pearson's coefficient of correlation: It is simple correlation and commonly used
 - Charles Spearman's coefficient of correlation
- In case of multivariate population; correlation can be studied through:
 - Coefficient of multiple correlation
 - Coefficient of partial correlation

- Is there any cause and effect (causal relationship) between two variables or between one variable on one side and two or more variables on the other side?
- This question can be answered by the use of regression analysis
- There are different techniques of regression.
 - In case of bivariate population; cause and effect relationship can be studied through simple regression.
 - In case of multivariate population; Causal relationship can be studied through multiple regression analysis.

Time series Analysis;

- Successive observations of the given phenomenon over a period of time are analyzed through time series analysis.
- It measures the relationship between variables and time (trend)
- The past trend can be used to evaluate the success or failure of management or any other policy.
- Based on past trend the future patterns can be predicted and policy may accordingly be formulated.



Chapter Nine

Interpretation and Reporting the Research Result

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- Interpretation refers to the task of drawing inferences from the collected facts after analytical or experimental study
 - Interpretation has to be done carefully so that misleading conclusion will not be drawn and the whole purpose of doing research will not be vitiated.
 - Researcher can only be better appreciated only through interpretation why his findings are, what they are and can make others to understand the real significance of his research findings.

Technique of interpretation

- The task of interpretation is not an easy job.
- Rather it requires a good skill on the part of researcher.
- Interpretation is an art that one learns through practice and experience.
- The researcher may, at times, seek the guidance from experts for accomplishing the task of interpretation.
- There are no existing rules to guide the researcher about how to interpret the data

- However, the following suggested steps could be helpful.
- Researcher must give reasonable explanation of the relation, which he has found and he must interpret the lines of relationship in terms of the underlying processes.
- Extraneous information, if collected during the study, must be considered while interpreting the final result of research study
- It is advisable, before embarking upon final interpretation, to consult someone having insight into the study and who is frank and honest and will not hesitate to point out omissions and errors in logical argumentation.
-
- Researcher must accomplish the task of interpretation only after considering all relevant factors affecting the problem to avoid false generalization

Precaution in interpretation

Researcher must pay attention to the following points for correct interpretation:

- At the outset, researcher must invariably satisfy himself that: the data are appropriate, trust worthy and adequate for drawing inferences
- The researcher must remain cautious about the errors that can possibly arise in the process of interpreting results
- Broad generalization must be avoided, because the coverage restricted to a particular time, a particular area and particular condition.
- The researcher must remember that there should be constant interaction between initial hypothesis and, empirical observation and theoretical conceptions

Reporting the Research Result

- Writing report is the last step in a research study and requires a set of skills somewhat different from those called for in research of the earlier stages of research
- This task should be accomplished by the researcher with at most care.
- He may also seek the assistance and guidance of experts for the purpose.
- Even the most brilliant hypothesis, well-designed and conducted research study, and the most striking generalization and findings are of little importance unless they are effectively communicated to others

Layout of Research Report

- Layout of the report means as to what the research report should contain and look like.
- A comprehensive layout of the research report should comprise:
 - Preliminary pages
 - The main text
 - The end matter

1) Preliminary pages

In this part the report should include:

- Title and Date
- Acknowledgment (this can be in the form of preface and forward, in larger study)
- Table of content
- List of tables (figures)

2) *Main text*

- The main text provides the complete outline of the research report along with all details.
- Title of the research is repeated at the top followed by abstract and then follows the other details on pages numbered consecutively beginning with second page.
- Each main section of the report should begin on a new page.

Main text can have the following sections

- Introduction
 - Background of the study
 - Rationale
- Objectives
- Literature Review
- Material and Methodology
 - Data (or material)
 - Methodology used,
 - Limitation of the study
- Results and discussion (in some cases, Empirical Analysis)
- Summary, Conclusion and Recommendation

3) End matter

- Here belong sections like:
 - **References (bibliography):** It should be based on alphabetical listing of names
 - **Appendix**

PRECAUTIONS FOR WRITING RESEARCH REPORTS

- While determining the length of the report, one should keep in view the fact that it should be long enough to cover the subject
- It should be such as to sustain reader's interest
- Abstract terminology and technical jargon should be avoided in a research report
- The report must provide a ready availability of the findings
- The layout of the report should be well thought out and must be appropriate

- The reports should be free from grammatical, punctuation, abbreviation and other mistakes.
- A research report should show originality and should necessarily be an attempt to solve some intellectual problem
- the report must also state the policy implications relating to the problem under consideration
- Report must be attractive in appearance, neat and clean, whether typed or printed



THANK YOU